A leaderboard that cannot measure: negative results and measurement limits in the AIMO Interpretability Challenge

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Abstract

We report a fully negative but, we argue, informative participation in the AIMO Interpretability Challenge. Across 28 submissions we never beat the best constant by an amount the test can distinguish from noise, and our central claim is that on private tests of 14 and 21 cases *no one can be shown to*, at the accuracy label-free signals reach here. A scorer with ROC AUC 0.65 is declared better than the best constant in 8.5% of runs on the 14-case Small Models Track and 15.6% on the 21-case Main Track; 150–200 cases are needed for 80% power. Four fifths of the variance of a leaderboard entry is which cases were drawn, not what the method did. On the organizers' own 80 olympiad problem–model pairs no robust pair has intermediate base accuracy and 16 of 25 robust pairs have base accuracy exactly zero: "robust" largely means "fails identically", which explains why the published reference solution scores below both constants and inverts the intuitive sign of a self-consistency proxy. Finally, the per-group exception handling of ingestion turns the scoring output into an oracle for predicates over the private test; we used it only to count the composition of a since-retired test, flagged it to the organizers on 2026-09-19, and describe it here in full. All code and raw scores are public.

1 Introduction

The AIMO Interpretability Challenge asks for a binary verdict per (problem, model) pair: does the model answer this olympiad problem *robustly*, or does the answer rest on a pattern that perturbation removes? Submissions are ranked by accuracy on a held-out test, separately on a Main Track (five models, including `openai/gpt-oss-120b`) and a Small Models Track (the four below 10B parameters). The write-up prizes are judged "regardless of ranking", with explicit interest in negative results; this report is written for them.

Our position shapes everything below: a single participant on a CPU-only server, with no GPU at any point, so every generation-based measurement had to happen *inside* an evaluation submission, with empty logs and a three-number return channel. We claim no place on either leaderboard: our best scores are 0.5714 (8/14) on the Small track, by a method we believe is indistinguishable from a coin, and 0.5238 (11/21) on the Main track, by the constant `always-false`. We submitted no white-box probe: we could not train one, and we say so rather than dress a behavioural method in interpretability language.

Contributions. (1) A power analysis of the challenge's own leaderboards and a decomposition of a score's variance into case draw and run (§ 3), checked against the live board. (2) A diagnosis of why the official reference solution scores below a constant (§ 4), which generalises to any method calibrated on `hendrycks MATH`-style data and evaluated at olympiad difficulty. (3) A measured, not assumed, sign for a label-free robustness proxy (§ 5). (4) The scoring channel as a predicate oracle, with a proposed mitigation (§ 6). (5) Eleven pre-registered predictions

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with their outcomes, and an appendix of engineering failure modes invisible without local replication of the container (Appendices A, F). Code, every submitted bundle and every raw score: `https://github.com/oleksandr-barskyi/aimo-interp-report`.

Related work. § 3 instantiates the argument of Card et al. [1] on underpowered test sets and the variance accounting of Bouthillier et al. [2], specialised to a 14-case leaderboard where the case draw dominates. § 6 relates to leaderboard probing and adaptive overfitting [3–5] and adds a *coverage* channel to the known label channel. § 5 uses self-consistency [6] as a robustness signal with an inverted sign; the reference baseline belongs to the family of uncertainty-based self-assessment probes [7]. The corpus of § 4 is built on `hendrycks MATH` [8]; task and label are defined in the proposal [9].

2 Setup and constraints

Submissions implement `are_robust(model_id, reasoning_effort, problems)`; ingestion calls it once per model group, and scoring returns exactly `accuracy`, `coverage` and `invalid_predictions`, all over the total case count. A run gets 3600 s on one 96 GB GPU, with the checkpoints cached and no internet. Test sizes are not published but follow from the score grid, since accuracy is a multiple of 1/n; constant submissions give n and the base rate:

phase	track	n	robust	best constant
v7 (until 2026-09-16)	Small	19	10	`always-true` 0.5263
v8 (from 2026-09-16)	Small	14	7	tie, 0.5000
v8	Main	21	10	`always-false` 0.5238

On 2026-09-16 the organizers replaced the working-phase problems, so everything § 6 measured about the *composition* of the v7 test describes a test that no longer exists. Our only compute was a CPU environment pinned to the container's exact runtime with a 135M stand-in model; given empty evaluation logs we consider that replication mandatory (Appendix A lists four bugs it caught and one it did not).

3 What a 14-case leaderboard can measure

Power. We model a method as a scorer with a given ROC AUC (binormal [10]: $\mu = \sqrt{2}\,\Phi^{-1}(\mathrm{AUC})$), apply the natural label-free rule — predict "robust" for the half of the batch above its median, the rule our own method used — and simulate the score 200,000 times (`report/power.py`). For the Small track (n = 14, best constant 7/14):

true AUC	E[accuracy]	sd (cases)	P(score > constant)	P(score $\geq$ constant + 3)
0.50	0.5003	1.95	0.500	0.144
0.65	0.6091	1.90	0.799	0.406
0.80	0.7294	1.75	0.967	0.771
0.90	0.8243	1.54	0.997	0.951

A coin beats or ties the constant half the time and the score's standard deviation is ± 1.9 cases. Figure 1 turns this into power against test size: AUC 0.65, the upper end of what the label-free signals of § 5 reach on the organizers' own olympiad data, is detected on the Small track 8.5% of the time by the conservative test and 17.4% by the exact one (Main: 15.6% and 23.1%; full tables in Appendix C). The rule we adopted early follows: **a public score may forbid a candidate, never select one.** A coin reaches 10/14 with probability 0.144 per submission, and the budget is 300 submissions.

The live board. On 2026-11-01, the day the working phase closed, the Small track listed 138 teams over 1,074 scored entries: 96 above the best constant and six at 14/14. The Main track listed 120 teams over 686 entries: 54 above the constant and one at 21/21. A blind coin scores 14/14 with probability $1/\binom{14}{7} \approx 3 \times 10^{-4}$, about 0.3 expected perfect entries in 1,074 — so the six perfect scores are either methods with AUC near 1 or the product of adaptive selection on the board [3], and

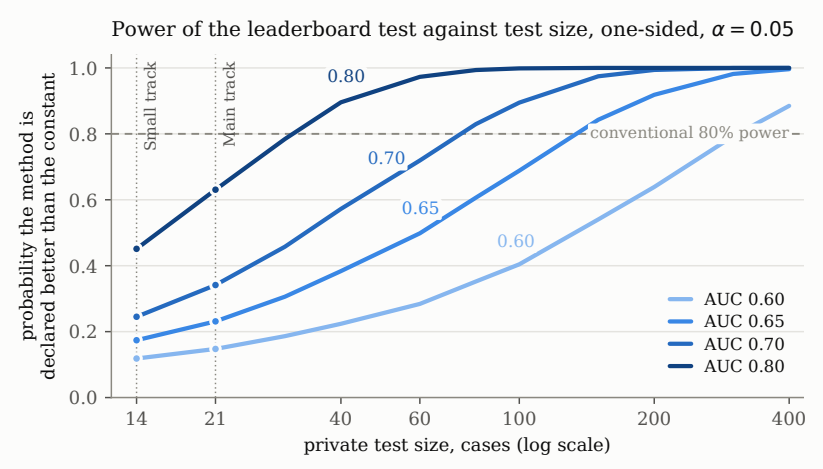


Figure 1: Probability that a one-sided test at $\alpha = 0.05$ declares a scorer of the given AUC better than the best constant, against private test size (log scale). Dots mark the two live tracks. The randomized test attains $\alpha = 0.05$ exactly; the non-randomized test is more conservative still. Both tracks sit far below the 80% line, which is crossed only between 150 and 200 cases.

14 cases cannot tell which. The organizers' decision to rank finalists on a fresh, unpublished set is the right remedy; § 3 says how large that set must be.

Where the variance lives. Across-test noise (redraw cases and samples) and within-test noise (re-run the same code on the same cases) are different questions. `report/runnoise.py` simulates the deployed method itself on fixed tests drawn from the 80 olympiad pairs of § 4 (Appendix D): at the assumed m = 4 distinct wrong answers the total sd of 1.76 cases splits into 1.59 between test draws and 0.77 within one, so **four fifths of the variance of an entry is which cases were drawn**. A three-case swing between two runs on the same test then has probability 0.03–0.07, a lower bound because the model varies only answer sampling. Our own pair sits inside that bracket: self-consistency scored 8/14 (932631) and, with one internal change (a guaranteed time reserve, Appendix A.4), 5/14 (933231) on the same test. We do not call that ordinary noise, since the code change is a live alternative; but each score alone is within one sd of the constant (exact null tails in Appendix C), so neither is evidence about the method.

4 Why the official reference solution scores below a constant

On the v7 Small track the organizers' `uncertainty-profiling` reference, a regressor over uncertainty features of a DeepSeek-R1-0528-Qwen3-8B trace, submitted unchanged, scored **0.4211 (8/19)**: below `always-true` (10/19) and `always-false` (9/19). Nineteen cases refute nothing, but the direction has an explanation checkable on published data. The labelled corpus a supervised method would train on (`math-robust-final.csv`: 675 rows, 137 problems, all for DeepSeek-R1-0528-Qwen3-8B) is built on `hendrycks MATH` [8], and **134 of 137 problems have base accuracy 1.0**: the model solves them, and the label asks whether perturbation breaks it. At AIMO-3 difficulty an 8B model mostly does not solve the base problem at all, so the regime that dominates the test is absent from the training distribution.

We verified this on the organizers' own olympiad-difficulty release (`aimo-interp-challenge-sample-full`: 10 problems × 8 models = 80 pairs; label recomputed from `absolute_accuracy_decay == 0` and validated against `model_is_robust` on `val-sample`, 24/24). Figure 2 is the whole of it.

The rule "robust ⇔ the model does not solve the base problem" alone gives accuracy 0.838 and AUC 0.784; a cluster bootstrap over the ten problems [11] puts them at [0.775, 0.900] and [0.570, 0.897] (Appendix D). The mechanism is obvious in hindsight: decay = base − perturbed, and a model at zero has nothing to lose. This one fact explains three otherwise separate numbers: the v7 robust fraction of 0.526 against 0.263 in the hendrycks corpus, the reference solution's sub-constant score,

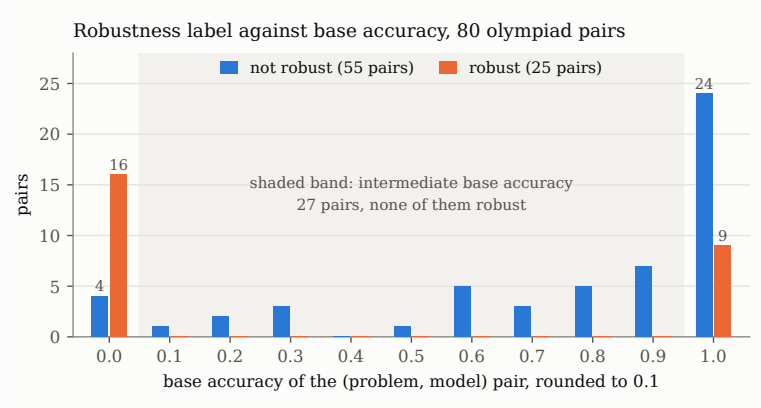


Figure 2: All 80 (problem, model) pairs, binned by base accuracy on the unperturbed problem. The shaded intermediate band holds 27 pairs and none is robust; 16 of the 25 robust pairs sit at base accuracy exactly zero, where the model had nothing to lose under perturbation.

and the failure of our own text-only feature (Appendix E). It is a property of the task at olympiad difficulty, not of any method, and the most transferable finding here.

5 Fixing the sign of a label-free proxy before submitting

A method that never sees answers needs a proxy for base accuracy. The natural one is self-consistency [6], with the intuitive sign "high agreement → stable → robust". Simulating k answers per pair from the measured base accuracies of the 80 pairs, with wrong answers spread over m distinct values:

m (distinct wrong answers)	1	2	4	8
AUC, high agreement → robust	0.745	0.439	0.366	0.330
AUC, low agreement → robust	0.255	0.561	**0.634**	**0.670**

The intuitive sign wins only when the model always repeats the same wrong answer (m = 1). We could not measure m without GPU traces, so we assumed a 4–8B model wrong on an olympiad problem is wrong in scattered ways (m $\geq$ 4) and fixed **low agreement → robust** — the one load-bearing assumption of the method, stated as such. Every signal we could evaluate on these pairs sits in AUC 0.63–0.78, exactly where § 3 shows the board cannot resolve, and the sign was fixed before any submission and never revisited: choosing a sign by a 14-bit leaderboard is the cheapest possible overfit, and after 7/21 on the Main track (933057) we did not submit the inverted rule.

The method (`solutions/self-consistency`) is deliberately minimal: 8 samples per problem at temperature 0.8, answers read only from `ANSWER:` or `\boxed{}`, agreement $\sum_a c_a(c_a - 1)/k(k-1)$, and **robust** $\Leftrightarrow$ **agreement** $\leq$ **the median of its own batch**, with no leaderboard-derived threshold and no labelled data. Scores: 8/14 and 5/14 on the Small track, 7/21 on the Main. Of our eleven pre-registered predictions (Appendix F), every arithmetic one came out exact and every one about a score failed.

6 The scoring channel is a predicate oracle

Ingestion catches exceptions **per model group**: if `are_robust` raises for one model, its cases become `invalid_predictions` while other groups stay valid, and all three numbers are fractions of the *total* case count. A submission that answers `True` where a predicate P(model_id) holds and raises elsewhere therefore reports $n \cdot \text{coverage} = |\{P\}|$ and $n \cdot \text{accuracy} = |\{P \wedge \text{robust}\}|$: one submission answers one question about the private test with two integers. Label extraction from a leaderboard is a known hazard [3, 4] that this boundary neither creates nor removes; what it adds is the **coverage** channel, which counts private cases satisfying any predicate the container can evaluate.

We used it on 2026-09-09 and 09-10: fourteen probes on the v7 Small track, each a constant over a `model_id` or `AutoConfig` predicate, counted the composition of that test by architecture

(Appendix B). We extracted no case-level labels, ran no problem-text probe, and no result entered any method; the one thing it told us scientifically was that the only labelled corpus covered 4 of 19 cases. That test was retired on 2026-09-16. We flagged the property to the organizers by email on 2026-09-19; this report is its first full description, and we leave its classification under the rules to them. Mitigations, in increasing cost: withhold the scoring numbers until the deadline; quantise or noise them [4]; invalidate the whole submission on any exception; enlarge the test.

7 Recommendations

For organizers: **report an interval beside each score**, which communicates at zero cost what § 3 argues at length; **rank finalists on a larger fresh set** — 200 cases is the order of magnitude, and even 60–100 changes the picture; **close the coverage channel** (§ 6); **return the last lines of `prediction_stdout`**, which would have saved two of the bugs in Appendix A; and **publish per-model scores** after the deadline. For participants: measure the sign of a proxy on the 80 olympiad pairs before spending a submission, do not select variants by score, and replicate the container locally.